Gabriela Chavgova

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Applied ML Scientist / AI Research Engineer | EU work authorization | Available Sep 2027

AI researcher and engineer experienced in multimodal learning, LLM post-training and deployment, diffusion models, mechanistic interpretability, scalable experimentation, and client-facing delivery.

Experience

Incoming Machine Learning Scientist Intern at TNO, The Hague **Jan 2027 - Jun 2027**

- MSc thesis on VLM/LLM-based agents for anomaly detection across multimodal surveillance and inspection data, including video, visitor records, and inspection logs.

Data Science Intern at ABN AMRO, Detecting Financial Crime, Amsterdam **Apr 2026 - Aug 2026**

- Independently framed and executed an end-to-end NLP research project for merchant extraction from noisy payment-service-provider transactions; implemented and benchmarked BiLSTM, Transformer, and BERTje models with CRF decoding, auxiliary objectives, and provider-balanced sampling.
- Built reproducible PyTorch/PySpark workflows for feature generation, multi-GPU training, experiment tracking, threshold selection, and unseen-provider evaluation using Databricks, Azure, and MLflow; delivered the final technical recommendation.

Application Software / AI Engineer at Thermo Fisher Scientific, part-time, Eindhoven **Feb 2025 - Apr 2026**

- Built and productionized an RTMDet vision pipeline for SEM imagery, integrating inference into Phenom microscope workflows and improving detection accuracy by 18% through targeted experimentation.
- Developed configurable multithreaded workflows for Phenom SEM systems, translating domain requirements into maintainable software components.

Scientific Mentor at Summer Research School in Informatics and Mathematics **Aug 2022, Aug 2023**

- Mentored seven international students on applied ML research, reviewed experiments and implementations, and helped one student qualify for Regeneron ISEF.

Education

University of Amsterdam (UvA), Master of Science in Artificial Intelligence **Sep 2025 - Jul 2027**

GPA: 7.8/10; coursework in Deep Learning, Computer Vision, NLP, Reinforcement Learning, and Information Retrieval.

Eindhoven University of Technology (TU/e), Bachelor of Science in Computer Science **Aug 2022 - Jul 2025**

GPA: 7.6/10; Honors Academy graduate, High Tech Systems track; coursework in algorithms, probability and statistics, computer networks, and security.

Projects and Research Experience

Mechanistic Interpretability and Identifiability in Neural Networks **2026**

- Reproduced and extended research on the non-identifiability of neural-network circuit explanations; developed a goal-relative framework covering six explanation properties.
- Extended single-neuron logical abstractions to distributed multi-neuron circuits and evaluated how stronger interpretability constraints change the number and stability of valid explanations.

Quantifying Creativity in Discrete Diffusion Models **2026**

- Adapted a continuous-diffusion creativity framework to D3PMs; implemented conditional ResNet/U-Net models with absorbing and uniform noise processes in PyTorch.
- Ran reproducible experiments on Snellius H100 GPUs, calibrated reverse-process scorers using per-step KL divergence, and evaluated fidelity with macro-F1, Cohen's κ, and token accuracy.

Automated Travel Information Extraction System - Industry Bachelor Project at TU/e **2025**

- Curated task-specific data and fine-tuned Llama 3.x with LoRA/QLoRA for schema-constrained extraction; deployed a Dockerized FastAPI service through `llama.cpp` on CPU/GPU servers.
- Implemented JSON-schema validation, tests, safeguards, confidence scoring, and hallucination checks, reducing client costs by 70%; worked with a ten-person team and client on delivery.

Achievements and Conferences

- **Regeneron ISEF 2021 Second Grand Award** for a CNN speech-emotion robot deployed on Raspberry Pi; three-time medal recipient at the Bulgarian Student Research Conference (2020-2022).
- Published in *Digital Transformation in the Post-Information Age* (ISBN 978-954-334-251-8); presenter at the 2021 Education and Research in the Information Society Conference.

Skills

Programming: Python, Java | **Languages:** Bulgarian (native), English (C1)
ML and research: PyTorch, TensorFlow/Keras, scikit-learn, LLM fine-tuning, VLMs, diffusion models, mechanistic interpretability, multimodal learning, NLP, computer vision, speech
Training and evaluation: multi-GPU training, HPC, ablation studies, model calibration, out-of-distribution evaluation, error analysis, hyperparameter optimization
Engineering: Databricks, PySpark, MLflow, Azure, Valohai, Docker, FastAPI, `llama.cpp`, Linux, Git, pytest, LaTeX